# Chapter 7
# Time-Varying Fields and Maxwell's Equations

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# Contents

# 7.1 Faraday's Law of Electromagnetic Induction

In 1820, Oersted: discovered magnetic effect on current.

In 1831, Faraday: discovered law of electromagnetic induction.

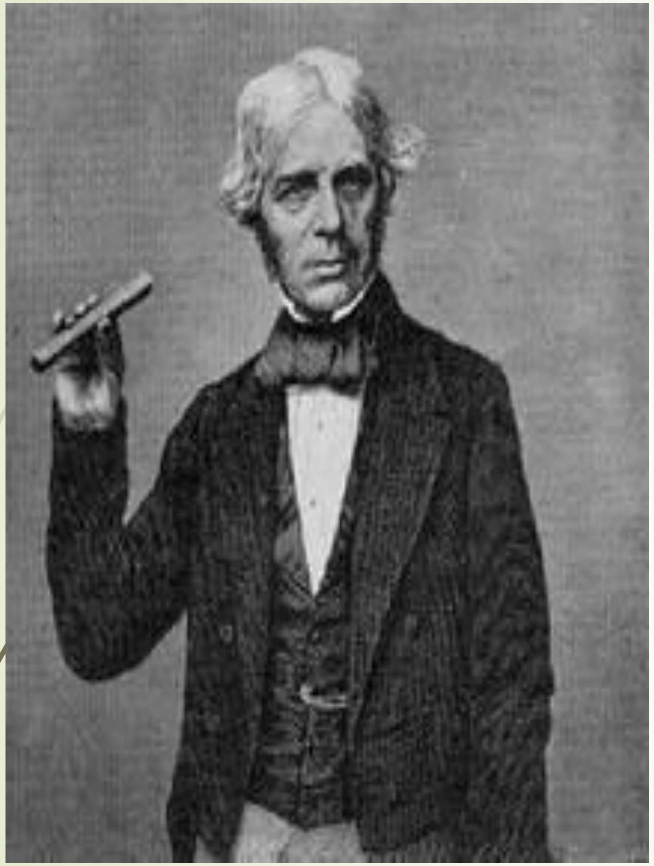
- Faraday's Law of electromagnetic induction

- Reveal the time varying magnetic field generate the electric field;

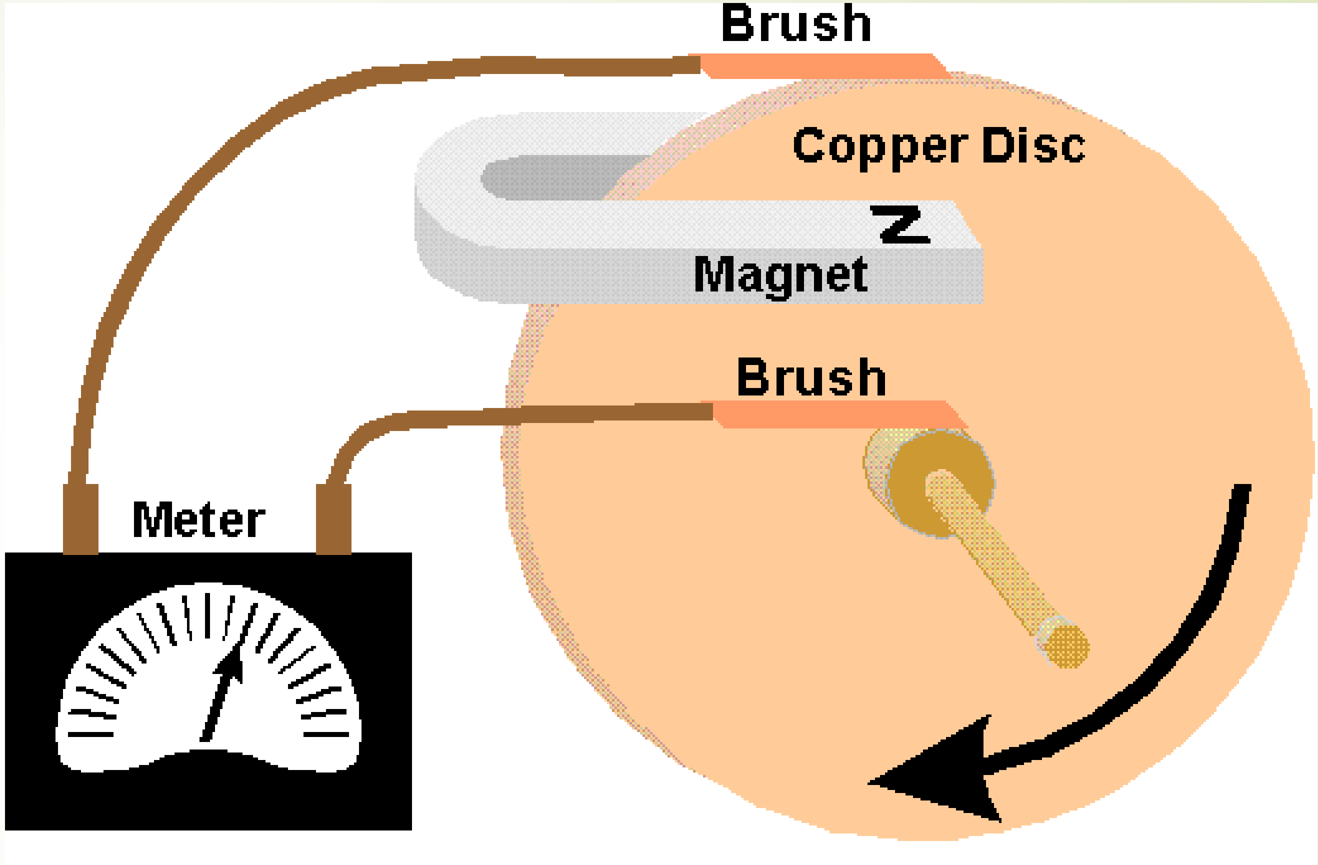
- It is an experimental law and can be considered as a postulate;

- Important conclusions：

- Under time-varying conditions, Electric field and magnetic field will be induced mutually by each other, forming a unified electromagnetic field .



Michael Faraday
22nd, Spt,1791—25th, Aug, 1867




Symbolic illustration of the Faraday Dynamo

Extend reading

## 法拉第发现电磁感应现象的背后

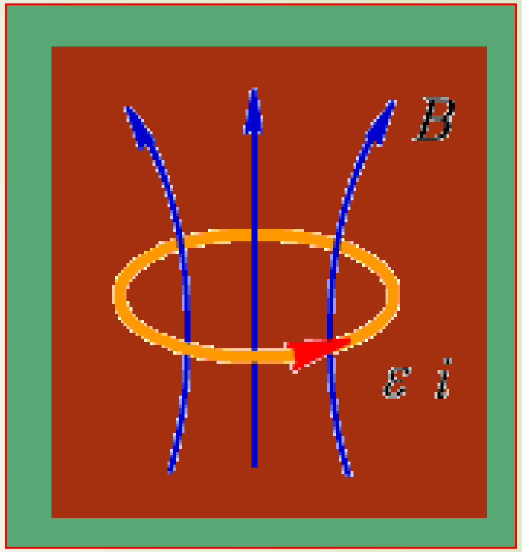
- 法拉第在1823年开始做了一系列实验，企图找到由磁生电的方法。法拉第年复一年地进行各种可能设想的实验，经历了无数次的失败，终于在1831年取得突破性进展。

| | | |
|---|---|---|
| 安培1822年发现了现象，但没深入研究 | 科拉顿的实验装置和法拉第几乎一样，但是把电流表放在另外一个房间。没观察到现象 | 历经8年，无数次失败终于发现了电磁感应现象。 |

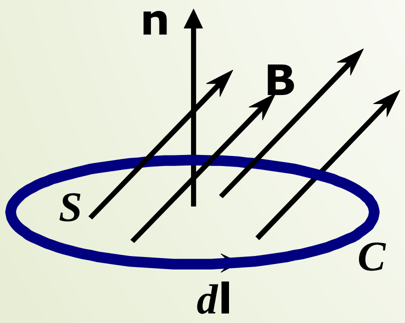
- 法拉第的学生和朋友丁铎尔（J.Tyndall）在法拉第传记中写道“这位铁匠的儿子，订书商的学徒，他的一生一方面是可以得到十五万英镑的财富，一方面是完全没有报酬的学问，要在这两者之间做出选择，结果他选择了后者，终生过着穷困的日子。然而，他却使英国的科学声誉比各国都高，获得接近四十年的光荣”

## 1. Expression of Faraday's law of electromagnetic induction

$$\varepsilon_{\text{in}} = -\frac{\mathrm{d}\Psi}{\mathrm{d}t}$$




- The electromotive force (EMF) induced in a stationary closed circuit is equal to the negative rate of increase of the magnetic flux linking the circuit.




- The negative sign

  - oppose the change--- **lenz's Law**

$$\Psi = \int_S \vec{B} \cdot \mathrm{d}\vec{S} \quad \Rightarrow \quad \varepsilon_{\text{in}} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_S \vec{B} \cdot \mathrm{d}\vec{S}$$

As: $\varepsilon_{in} = \oint_C \vec{E}_{in} \cdot \mathrm{d}\vec{l}$

thus: $\oint_C \vec{E}_{in} \cdot \mathrm{d}\vec{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_S \vec{B} \cdot \mathrm{d}\vec{S}$

- **Understanding of induced electric field :**

• **The changing magnetic field is the source of the electric field.**

• **The induced electric field is a rotational field.**

**The sources of the electric field have two kinds: Charge, Magnetic field (change with time)**

$$\vec{E} = \vec{E}_{\mathrm{in}} + \vec{E}_{\mathrm{c}} \qquad \oint_C \vec{E}_c \bullet \mathrm{d}\vec{l} = 0$$

$$\oint_C \vec{E} \bullet \mathrm{d}\vec{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_S \vec{B} \bullet \mathrm{d}\vec{S}$$

**2. Several conditions made magnetic flux change in the loop**

**(1) Magnetic field changes with time while Loop is invariant.**

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_S \vec{B} \cdot \mathrm{d}\vec{S} = \int_S \frac{\partial \vec{B}}{\partial t} \cdot \mathrm{d}\vec{S} \Rightarrow \oint_C \vec{E} \bullet \mathrm{d}\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \bullet \mathrm{d}\vec{S}$$

**The corresponding differential form is**

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

## ( 2 ) The motion of a conductor loop in a constant magnetic field

$$\varepsilon_{in} = \int_C \vec{E} \cdot \mathrm{d}\vec{l} = \int_C (\vec{v} \times \vec{B}) \cdot \mathrm{d}\vec{l}$$

**Which is called motional electromotive force , this is the working principle of the generator.**

## ( 3 ) The loop moves in a time varying magnetic field

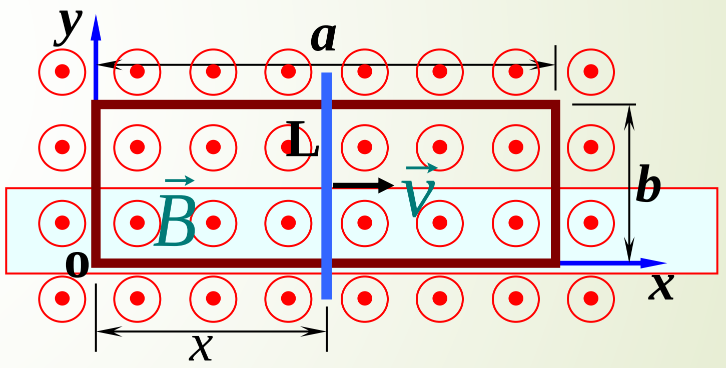
$$\varepsilon_{in} = \int_C \vec{E} \cdot \mathrm{d}\vec{l} = \int_C (\vec{v} \times \vec{B}) \cdot \mathrm{d}\vec{l} - \int_S \frac{\partial \vec{B}}{\partial t} \cdot \mathrm{d}\vec{S}$$

**Ex.** A uniform magnetic field B vertical through a rectangular ring with a length of *a* and a width of *b* , as shown in figure. In the following three cases, solve the induction electromotive force .

(1) $\vec{B}=\vec{e}_z B_0\cos(\omega t)$ , Rectangular loop is static ;

(2) $\vec{B}=\vec{e}_z B_0$ , the width *b* of rectangular loop equals a constant, but the long side increases with time because sliding conductor L moves with velocity $\vec{v}=\vec{e}_x v$ ;

(3) $\vec{B}=\vec{e}_z B_0\cos(\omega t)$ , and sliding conductor L in rectangular loop moves with velocity $\vec{v}=\vec{e}_x v$ .



Rectangular ring in uniform magnetic field

Solution: (1) Uniform magnetic field $\vec{B}$ do simple harmonic change with time, and loop is static , thus induction electromotive force in the loop is generated by the change of magnetic field, so:

$$\varepsilon_{\text{in}}=-\int_S \frac{\partial \vec{B}}{\partial t}\cdot \mathrm{d}\vec{S}=-\int_S \frac{\partial}{\partial t}[\vec{e}_z B_0\cos(\omega t)]\cdot \vec{e}_z \mathrm{d}S=\omega a b B_0\sin(\omega t)$$

( 2 ) Uniform magnetic field $\vec{B}$ is constant magnetic field, the sliding conductor on the loop do uniform motion at speed of , thus the induction electromotive force in the loop is generated by the movement of conductor L in the magnetic field, so

$$\varepsilon_{in} = \int_C (\vec{v} \times \vec{B}) \cdot \mathrm{d}\vec{l} = \int_C (\vec{e}_x v \times \vec{e}_z B_0) \cdot \vec{e}_y \mathrm{d}l = -vB_0 b$$

or

$$\varepsilon_{\mathrm{in}} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_S \vec{B} \cdot \mathrm{d}\vec{S} = -\frac{\mathrm{d}}{\mathrm{d}t}(bB_0 vt) = -bB_0 v$$

( 3 ) The induced electromotive force in the rectangular circuit is produced by the magnetic field variation and the movement of the sliding conductor L in the magnetic field, so

$$\varepsilon_{in} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot \mathrm{d}\vec{S} + \oint_C (\vec{v} \times \vec{B}) \cdot \mathrm{d}\vec{l}$$

$$= \int_S \frac{\partial}{\partial t}[\vec{e}_z B_0 \cos(\omega t)] \bullet \vec{e}_z \mathrm{d}S + \oint_C [\vec{e}_x v \times \vec{e}_z B_0 \cos(\omega t)] \cdot \vec{e}_y \mathrm{d}l$$

$$= vt\omega bB_0 \sin(\omega t) - vbB_0 \cos(\omega t)$$

# 7.2 Maxwell's Equations

**Static state :** $\dfrac{\partial}{\partial t} = 0$ **Time varying state :** $\dfrac{\partial}{\partial t} \neq 0$

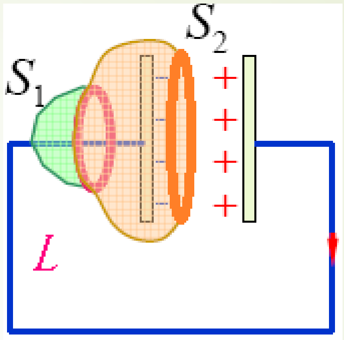
$\nabla \times \vec{E} = 0$ $\nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$ $\dfrac{\partial \vec{B}}{\partial t} \Rightarrow \vec{E}$

$\nabla \cdot \vec{J} = 0$ $\nabla \cdot \vec{J} = -\dfrac{\partial \rho}{\partial t}$

$\nabla \times \vec{H} = \vec{J}$ $\nabla \times \vec{H} = ?$ $\dfrac{\partial \vec{E}}{\partial t} \Rightarrow \vec{B}\ ?$

## Discussion on Ampere’s circuiltal Law

$$\oint_c \vec{H} \bullet d\vec{l} = \int_s \vec{J} \bullet d\vec{S} = I$$




**For S1:** $\oint_c \vec{H} \bullet d\vec{l} = \int_{S_1} \vec{J} \bullet d\vec{S} = I$

**For S2:** $\oint_c \vec{H} \bullet d\vec{l} = \int_{S_2} \vec{J} \bullet d\vec{S} = 0$

**Disagreement!**

**Conclusion**: midification should be made to Ampere’s circuital Law

## 1. Total current law

**Under the condition of time-varying situation, the charge distribution changes with time, according to current continuity equation, we got**

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \neq 0$$

**Appear contradictory**

**As** $\nabla \times \vec{H} = \vec{J}$ $\Rightarrow$ $\nabla \cdot \vec{J} = \nabla \cdot (\nabla \times \vec{H}) = 0$

**Not applied in the case of time-varying**

**Is there other curl source of magnetic field?**

$$\nabla \cdot \vec{D} = \rho \Rightarrow \nabla \cdot \vec{J} = -\frac{\partial}{\partial t}(\nabla \cdot \vec{D}) \Rightarrow \nabla \cdot (\vec{J} + \frac{\partial \vec{D}}{\partial t}) = 0$$

**Modify** $\nabla \times \vec{H} = \vec{J}$ **to** $\nabla \times \vec{H} = \vec{J} + \dfrac{\partial \vec{D}}{\partial t}$ ☞ **Conflict resolution**

**The time varying electric field will give rise to a magnetic field**

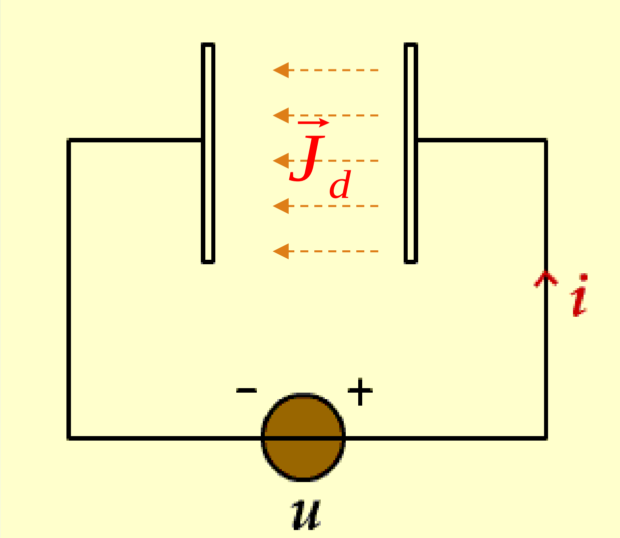
## Total current law :

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$ —— **Differential Form**

$$\oint_C \vec{H} \cdot \mathrm{d}\vec{l} = \int_s (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \cdot \mathrm{d}\vec{S}$$ —— **Integral Form**

**Total current law reveals that not only the <u>conduction current</u> can induce magnetic field, but also the <u>time-varying electric field</u> can give rise to a magnetic field. It forms an exchange in the nature between the time-varying magnetic field and the time-varying electric field.**

## 2. Displacement current density




$$\vec{J}_{\mathrm{d}} = \frac{\partial \vec{D}}{\partial \mathrm{t}}$$

- **The rate of change over time of electric displacement vector can produce magnetic field, just like current, so it is called "displacement current”.**
- **The displacement current only indicates the rate of change of the electric field, which is different from the conduction current, and it has no heat effect.**
- **The introduction of the displacement current is a vital step in the establishment of Maxwell equations. It reveals the important physical concept of the time varying electric field generate magnetic field.**

### Attention：

**• In an insulating medium, there is no conduction current, but there is a displacement current.**
**• In an ideal conductor, there is no displacement current, but there is a conduction current.**
**•In the general medium, there are both the conduction current and the displacement current.**

## 3.Maxwell equations —— Basic equations of electromagnetic field

**Differential form**

$$\begin{cases} \nabla \times \vec{\mathbf{H}} = \vec{\mathbf{J}}_f + \dfrac{\partial \vec{\mathbf{D}}}{\partial t} \\ \nabla \times \vec{\mathbf{E}} = -\dfrac{\partial \vec{\mathbf{B}}}{\partial t} \\ \nabla \cdot \vec{\mathbf{B}} = 0 \\ \nabla \cdot \vec{\mathbf{D}} = \rho_f \end{cases}$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

**Integral form**

$$\begin{cases} \oint_C \vec{H} \cdot \mathrm{d}\vec{l} = \int_S (\vec{J}_f + \dfrac{\partial \vec{D}}{\partial t}) \cdot \mathrm{d}\vec{S} \\ \oint_C \vec{E} \cdot \mathrm{d}\vec{l} = -\int_S \dfrac{\partial \vec{B}}{\partial t} \cdot \mathrm{d}\vec{S} \\ \oint_S \vec{B} \cdot \mathrm{d}\vec{S} = 0 \\ \oint_S \vec{D} \cdot \mathrm{d}\vec{S} = \int_V \rho_f \, dV \end{cases}$$

$$\oint_S \vec{\mathbf{J}} \cdot \mathrm{d}\vec{\mathbf{S}} = -\frac{d}{dt} q$$

## 4. Constitutive relation of medium

**The constitutive relation of an isotropic linear medium is**

$$\vec{D} = \varepsilon \vec{E} \quad \vec{B} = \mu \vec{H} \quad \vec{J} = \sigma \vec{E}$$

**Substitute into the Maxwell equations， we got**

$$\begin{cases} \nabla \times \vec{H} = \sigma \vec{E} + \dfrac{\partial}{\partial t}(\varepsilon \vec{E}) \\ \nabla \times \vec{E} = -\dfrac{\partial}{\partial t}(\mu \vec{H}) \\ \nabla \cdot (\mu \vec{H}) = 0 \\ \nabla \cdot (\varepsilon \vec{E}) = \rho \end{cases}$$

**Restricted form of Maxwell equation**

（ **Homogeneous medium** ）

$$\begin{cases} \nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \dfrac{\partial \vec{E}}{\partial t} \\ \nabla \times \vec{E} = -\mu \dfrac{\partial \vec{H}}{\partial t} \\ \nabla \cdot \vec{H} = 0 \\ \nabla \cdot \vec{E} = \rho / \varepsilon \end{cases}$$

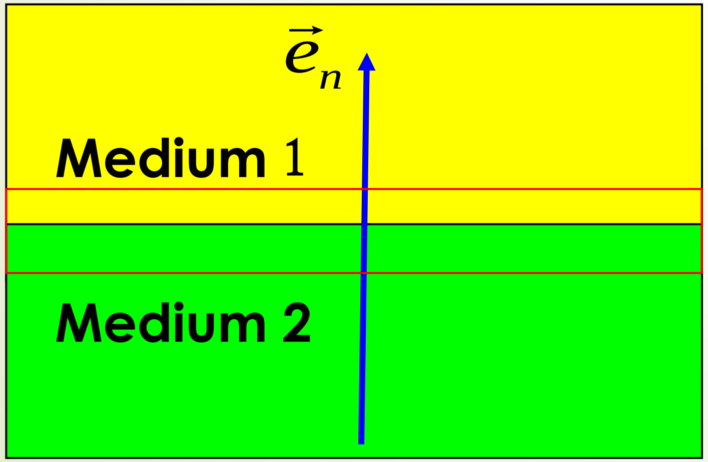
# 7.3 Electromagnetic Boundary Conditions

- **When letting the height of the flat closed path approach zero, the area bounded by the path approaches zero, causing the surface integrals of $\frac{\partial \vec{B}}{\partial t}$ and $\frac{\partial \vec{D}}{\partial t}$ to vanish;**

- **So the boundary conditions for time-varying case are exactly the same as for static electric fields and for static magnetic fields;**

## 1. The general expressions of boundary conditions

$$\begin{cases} \oint_C \vec{H} \cdot \mathrm{d}\vec{l} = \int_S (\vec{J} + \dfrac{\partial \vec{D}}{\partial t}) \cdot \mathrm{d}\vec{S} \\ \oint_C \vec{E} \cdot \mathrm{d}\vec{l} = -\int_S \dfrac{\partial \vec{B}}{\partial t} \cdot \mathrm{d}\vec{S} \\ \oint_S \vec{B} \cdot \mathrm{d}\vec{S} = 0 \\ \oint_S \vec{D} \cdot \mathrm{d}\vec{S} = \int_V \rho dV \end{cases}
\Rightarrow
\begin{cases} \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_S \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_S \end{cases}$$




The free surface charge density on the interface

The free surface current density on the interface

## 2. The boundary conditions of ideal dielectric interface

**that is $J_S = 0$、$\rho_S = 0$， so**

$$\begin{cases} \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = 0 \\ \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = 0 \end{cases}$$

- The tangential component of $\vec{D}$ is continuous
- The tangential component of $\vec{B}$ is continuous
- The tangential component of $\vec{E}$ is continuous
- The tangential component of $\vec{H}$ is continuous

## 3. The boundary conditions on the surface of the ideal conductor

$$\begin{cases} \vec{e}_n \cdot \vec{D} = \rho_S \\ \vec{e}_n \cdot \vec{B} = 0 \\ \vec{e}_n \times \vec{E} = 0 \\ \vec{e}_n \times \vec{H} = \vec{J}_S \end{cases}$$

- **The charge density of ideal conductor on the surface is equal to the normal component of $\vec{D}$**
- **The normal component of $\vec{B}$ on the surface of ideal conductor equals zero**
- **The tangential component of $\vec{E}$ on the surface of ideal conductor equals zero**
- **The current density of ideal conductor on the surface is equal to the tangential component of $\vec{H}$**

***E2、D2、H2、B2* all equal to zero**

# Electromagnetic Field

**Starting point**

**Maxwell's equations**

$$\begin{cases} \nabla \times \vec{H} = \vec{J} + \dfrac{\partial \vec{D}}{\partial t} \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{D} = \rho \end{cases}$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

**Conditions**

**Constitutive relations**

$$\begin{cases} \vec{D} = \varepsilon \vec{E} \\ \vec{B} = \mu \vec{H} \\ \vec{J} = \sigma \vec{E} \end{cases}$$

**Boundary conditions**

$$\begin{cases} \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_S \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_S \end{cases}$$

$$\vec{e}_n \cdot (\vec{J}_1 - \vec{J}_2) = -\frac{\partial}{\partial t} \rho_S$$

# 7.4 Wave Equations

$$\begin{cases} \nabla\times\vec{H} = \vec{J} + \varepsilon\dfrac{\partial\vec{E}}{\partial t} \\ \nabla\times\vec{E} = -\mu\dfrac{\partial\vec{H}}{\partial t} \\ \nabla\cdot\vec{H} = 0 \\ \nabla\cdot\vec{E} = \dfrac{1}{\varepsilon}\rho \end{cases} \Longrightarrow \nabla\times\nabla\times\vec{H} = \nabla\times\vec{J} + \nabla\times\left(\varepsilon\frac{\partial\vec{E}}{\partial t}\right)$$

$$\Downarrow$$

$$\nabla(\nabla\cdot\vec{H}) - \nabla^2\vec{H} = -\mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2} + \nabla\times\vec{J}$$

$$\Downarrow$$

$$\boxed{\nabla^2\vec{H} - \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2} = -\nabla\times\vec{J}}$$

**Similarly , we can get that**

$$\nabla^2\vec{E} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = \mu\frac{\partial\vec{J}}{\partial t} + \frac{1}{\varepsilon}\nabla\rho$$

## Equations of electric field and magnetic field

- Maxwell's equations ⇨ Wave equations:

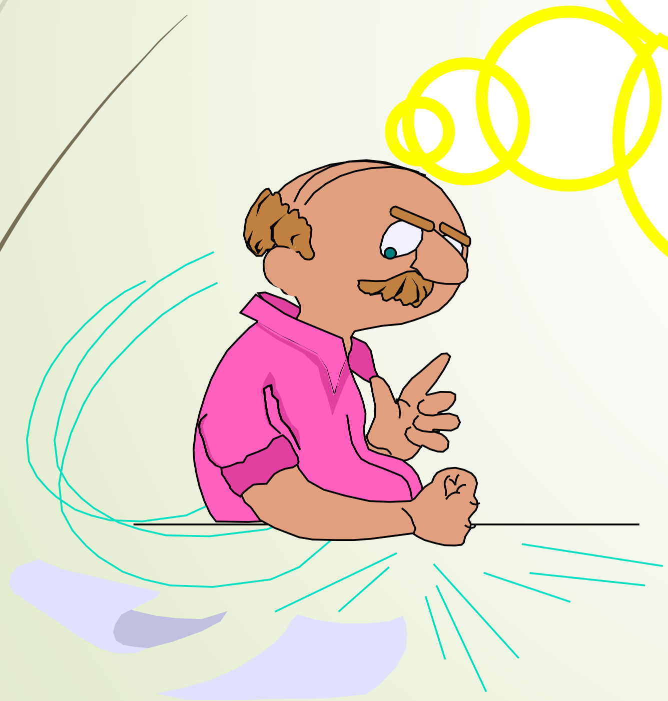
$$\nabla^2\vec{E}-\mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}=\mu\frac{\partial\vec{J}}{\partial t}+\frac{1}{\varepsilon}\nabla\rho$$

$$\nabla^2\vec{H}-\mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}=-\nabla\times\vec{J}$$

- Time-varying electric field and magnetic field exist in the form of waves

## Wave equation of source-free region

$$\nabla^2\vec{E}-\mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}=0 \qquad \nabla^2\vec{H}-\mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}=0$$



**Face the problems :**

- **Single medium environment!**
- **Solution of wave equations!**

**Analysis methods:**

- **Make full use the characteristics of time varying electromagnetic field.**

# Questions :

- In the electrostatic field analysis ,
  how to find its characteristics?
- How about in the static magnetic field?
- In the time-varying electromagnetic field analysis, will it play the same role?

# 7.5 Potential Functions

- **The introduction of potential functions**

$$\nabla \cdot \vec{B} = 0 \quad \Rightarrow \quad \vec{B} = \nabla \times \vec{A}$$

**Characteristic**

**The divergence of the magnetic vector potential can be arbitrary!**

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \Rightarrow \quad \nabla \times (\vec{E} + \frac{\partial \vec{A}}{\partial t}) = 0 \quad \Rightarrow \quad \vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla \varphi$$

- **Calculation of electromagnetic field by means of potential function**

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} - \nabla \varphi$$

**Questions:**

- **How to calculation a potential functions ?**
- **Is it beneficial to calculate the electromagnetic field by means of potential functions?**

- **Potential function equations （ D'Alembert Equations ）**

$$\nabla^2 \vec{A} - \varepsilon\mu \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J} + \nabla\left(\nabla \cdot \vec{A} + \mu\varepsilon \frac{\partial \phi}{\partial t}\right)$$

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon} - \frac{\partial}{\partial t}(\nabla \cdot \vec{A})$$

## Proof

$$\boxed{\vec{D} = \varepsilon \vec{E} \quad \vec{H} = \frac{\vec{B}}{\mu}}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \qquad \nabla \times \vec{B} = \mu \vec{J} + \varepsilon\mu \frac{\partial \vec{E}}{\partial t}$$

$$\boxed{\vec{B} = \nabla \times \vec{A} \quad \vec{E} = -\frac{\partial \vec{A}}{\partial t} - \nabla \phi}$$

$$\nabla \times \nabla \times \vec{A} = \mu \vec{J} + \varepsilon\mu \frac{\partial}{\partial t}\left(-\frac{\partial \vec{A}}{\partial t} - \nabla \phi\right)$$

$$\boxed{\nabla \times \nabla \times \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}}$$

$$\nabla^2 \vec{A} - \varepsilon\mu \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J} + \nabla\left(\nabla \cdot \vec{A} + \mu\varepsilon \frac{\partial \phi}{\partial t}\right)$$

# Similarly

$$\nabla\cdot\vec{D}=\rho$$

$$\boxed{\vec{D}=\varepsilon\vec{E}、\quad\vec{E}=-\frac{\partial\vec{A}}{\partial t}-\nabla\phi}$$

$$\varepsilon\nabla\cdot\left(-\frac{\partial\vec{A}}{\partial t}-\nabla\phi\right)=\rho$$

$$\nabla^2\varphi=-\frac{\rho}{\varepsilon}-\frac{\partial}{\partial t}(\nabla\cdot\vec{A})$$

## Potential function equations

$$\nabla^2 \vec{A} - \varepsilon\mu \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J} + \nabla\left(\nabla \cdot \vec{A} + \mu\varepsilon \frac{\partial \phi}{\partial t}\right)$$

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon} - \frac{\partial}{\partial t}(\nabla \cdot \vec{A})$$

**Disadvantages：**

**The scalar potential function and the vector potential functions can not be separated!**

## Normal conditions of potential functions

**Coulomb's gauge :** $\nabla \cdot \vec{A} = 0$

$$\nabla^2 \vec{A} - \varepsilon\mu \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J} + \mu\varepsilon \frac{\partial}{\partial t} \nabla \varphi$$

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon}$$

**Disadvantages ：**

**The scalar potential function and the vector potential functions can not be separated!**

## Normal conditions of potential functions

**Lorentz's gauge：** $$\nabla \cdot \vec{A} + \mu\varepsilon \frac{\partial \varphi}{\partial t} = 0$$

$$\nabla^2 \vec{A} - \varepsilon\mu \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J} \qquad \nabla^2 \phi - \varepsilon\mu \frac{\partial^2 \phi}{\partial t^2} = -\frac{\rho}{\varepsilon}$$

**Features ：**

- **The electric potential function and the magnetic vector potential function can be separated!**
- **The magnetic vector potential only depends on the current, and the electric potential function only depends on the charge!**
- **The potential functions can be found directly by the Lorentz's gauge (no need to solve the differential equations)!**

# Question :

- What is the role of the potential functions in the time-varying electromagnetic field ？

## Wave equations of electromagnetic field

$$\nabla^2 \vec{E} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} = \mu \frac{\partial \vec{J}}{\partial t} + \frac{1}{\varepsilon}\nabla\rho$$

$$\nabla^2 \vec{H} - \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2} = -\nabla \times \vec{J}$$

## Potential functions

$$\nabla^2 \vec{A} - \varepsilon\mu \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J}$$

$$\frac{\partial \varphi}{\partial t} = -\frac{1}{\mu\varepsilon}\nabla \cdot \vec{A}$$

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} - \nabla\varphi$$

### Conclusion :

- The two methods is just as simple in the source-free region
- The potential equations are much simpler in the source region

# 7.6 Time-Harmonic Fields

- **Concept of time-harmonic electromagnetic fields**

**The physical quantity changes with the time in accordance with the sine law, so it is also called the sine electromagnetic field problem.**

$$A(\vec{r},t) = A_0 \cos[\omega t + \phi(\vec{r})]$$

$$A(\vec{r},t) = A_0 \sin[\omega t + \phi(\vec{r})]$$

## The favorable factors for solving the problems of time-harmonic electromagnetic fields

**Time and space can be Separated to solution！**

**That is： It can be independent of each other to analyze the physical quantity of spatial variation and temporal variation.**

**Method：**

**The time-space relation of physical quantity can be expressed by the complex number**

$$A(\vec{r},t) = A_0(\vec{r})\cos(\omega t + \phi(\vec{r}) + \phi_0)$$
$$= \mathrm{Re}\left\{A_0 \mathrm{e}^{\mathrm{j}\phi_0}\mathrm{e}^{\mathrm{j}\phi(\vec{r})}\mathrm{e}^{\mathrm{j}\omega t}\right\}$$
$$= \mathrm{Re}[A(\vec{r})\mathrm{e}^{\mathrm{j}\omega t}]$$

**Methods of Real number representation or instantaneous representation**

$$A(\vec{r},t) \longleftrightarrow A(\vec{r})e^{j\omega t}$$

**Method of complex representation**

**When calculate the time-harmonic quantity about time derivative and integral , the complex representation is more advantageous**

$$A(\vec{r},t) = \mathrm{Re}[A(\vec{r})e^{j\omega t}]$$

| **Derivative** | **Integral** |
|---|---|
| $\frac{\partial}{\partial t}A(\vec{r},t) = \mathrm{Re}[j\omega A(\vec{r})e^{j\omega t}]$ | $\int dt A(\vec{r},t) = \mathrm{Re}[\frac{1}{j\omega}A(\vec{r})e^{j\omega t}]$ |
| $\frac{\partial}{\partial t}A(\vec{r},t) \longleftrightarrow j\omega A(\vec{r})$ | $\int A(\vec{r},t) \longleftrightarrow \frac{1}{j\omega}A(\vec{r})$ |
| $\frac{\partial}{\partial t} \longleftrightarrow j\omega$ | $\int dt \longleftrightarrow \frac{1}{j\omega}$ |

## The complex representation of Maxwell's equations
## ——Complex vector Maxwell's equations

$$\begin{cases} \nabla\times\vec{\boldsymbol{H}}(\vec{r},t)=\vec{\boldsymbol{J}}(\vec{r},t)+\dfrac{\partial\vec{\boldsymbol{D}}(\vec{r},t)}{\partial t} \\ \nabla\times\vec{\boldsymbol{E}}(\vec{r},t)=-\dfrac{\partial\vec{\boldsymbol{B}}(\vec{r},t)}{\partial t} \\ \nabla\cdot\vec{\boldsymbol{B}}(\vec{r},t)=0 \\ \nabla\cdot\vec{\boldsymbol{D}}(\vec{r},t)=\rho(\vec{r},t) \end{cases} \Longleftrightarrow \begin{cases} \nabla\times\vec{H}(\vec{r})=\vec{J}(\vec{r})+\mathrm{j}\omega\vec{D}(\vec{r}) \\ \nabla\times\vec{E}(\vec{r})=-\mathrm{j}\omega\vec{B}(\vec{r}) \\ \nabla\cdot\vec{D}(\vec{r})=\rho(\vec{r}) \\ \nabla\cdot\vec{B}(\vec{r})=0 \end{cases}$$

$$\nabla\cdot\vec{J}(\vec{r},t)=-\frac{\partial\rho(\vec{r},t)}{\partial t} \qquad \nabla\cdot\vec{J}(\vec{r})=-j\omega\rho(\vec{r})$$

## Complex representation of wave equations in the passive region ——Helmholtz’s equations

$$\frac{\partial}{\partial t} \to \mathrm{j}\omega \qquad \frac{\partial^2}{\partial t^2} \to -\omega^2$$

**Instantaneous vector**

$$\begin{cases} \nabla^2 \vec{E} - \mu\varepsilon \dfrac{\partial^2 \vec{E}}{\partial^2 t} = 0 \\ \nabla^2 \vec{H} - \mu\varepsilon \dfrac{\partial^2 \vec{H}}{\partial^2 t} = 0 \end{cases}$$

**Complex vector**

$$\begin{cases} \nabla^2 \vec{E} + k^2 \vec{E} = 0 \\ \nabla^2 \vec{H} + k^2 \vec{H} = 0 \end{cases}$$

$$(k = \omega\sqrt{\mu\varepsilon})$$

## The complex representation of the potential function equations in the active region

**Instantaneous vector**

$$\begin{cases} \nabla^2 \vec{\boldsymbol{A}} - \mu\varepsilon \dfrac{\partial^2 \vec{\boldsymbol{A}}}{\partial t^2} = -\mu \vec{\boldsymbol{J}} \\ \nabla^2 \varphi - \mu\varepsilon \dfrac{\partial^2 \varphi}{\partial t^2} = -\dfrac{\rho}{\varepsilon} \end{cases}$$

**Complex vector**

$$\begin{cases} \nabla^2 \dot{\vec{\boldsymbol{A}}} + k^2 \dot{\vec{\boldsymbol{A}}} = -\mu \dot{\vec{\boldsymbol{J}}} \\ \nabla^2 \varphi + k^2 \varphi = -\dfrac{\rho}{\varepsilon} \end{cases}$$

**Lorentz's condition**

$$\nabla \cdot \vec{\boldsymbol{A}} = -\mu\varepsilon \frac{\partial \varphi}{\partial t} \quad \Rightarrow \quad \nabla \cdot \vec{\boldsymbol{A}} = -\mathrm{j}\omega\mu\varepsilon\varphi$$

**Field based on potential functions**

$$\vec{B} = \nabla \times \vec{A}, \quad \vec{E} = -\frac{\partial \vec{A}}{\partial t} - \nabla\varphi$$

$$\Rightarrow \quad \vec{B} = \nabla \times \vec{A}, \quad \vec{E} = -j\omega\vec{A} - j\frac{\nabla\nabla\cdot\vec{A}}{\omega\mu\varepsilon}$$

## The simple representation of conductive medium——Complex dielectric constant

$$\nabla\times\vec{H}(\vec{r})=\vec{J}_{\text{introduced}}(\vec{r})+\vec{J}_{\sigma}(\vec{r})+\mathrm{j}\omega\vec{D}(\vec{r})$$

$$\vec{J}(\vec{r})=\sigma\vec{E}(\vec{r}) \qquad \vec{D}(\vec{r})=\varepsilon\vec{E}(\vec{r})$$

$$\nabla\times\vec{H}(\vec{r})=\vec{J}_{\text{introduced}}(\vec{r})+\sigma\vec{E}(\vec{r})+\mathrm{j}\omega\varepsilon\vec{E}(\vec{r})$$

$$\nabla\times\vec{H}(\vec{r})=\vec{J}_{\text{introduced}}(\vec{r})+\mathrm{j}\omega\left(\varepsilon-j\frac{\sigma}{\omega}\right)\vec{E}(\vec{r})$$

$$\nabla\times\vec{H}(\vec{r})=\vec{J}_{\text{introduced}}(\vec{r})+\mathrm{j}\omega\varepsilon_{c}\vec{E}(\vec{r})$$

## Medium loss tangent

For electromagnetic wave, the imaginary part of the complex permittivity of the medium will make the attenuation of the electromagnetic wave。 Therefore, in terms of electromagnetic wave propagation, dielectric constant / magnetic permeability with imaginary part of the medium is also called loss medium,

$$\varepsilon_c = \varepsilon^{'} - j\varepsilon^{''}$$

**The ratio of the real and imaginary parts, called loss tangent :**

**Dielectric** $\tan\delta_\varepsilon = \dfrac{\varepsilon''}{\varepsilon'}$, **Magnetic** $\tan\delta_\mu = \dfrac{\mu''}{\mu'}$, **Conductive medium** $\tan\delta_\sigma = \dfrac{\sigma}{\omega\varepsilon}$

Classification of conductive medium

$\dfrac{\sigma}{\omega\varepsilon} << 1$ —— **Weakly conducting medium and good insulator**

$\dfrac{\sigma}{\omega\varepsilon} \approx 1$ —— **Normal conducting medium**

$\dfrac{\sigma}{\omega\varepsilon} >> 1$ —— **Good conductor**

**The Ratio of conduction current and displacement current**

## Complex representation of wave equation in lossy medium

Source-free region

$$\begin{cases} \nabla^2 \vec{\boldsymbol{E}} + k_c^2 \vec{\boldsymbol{E}} = 0 \\ \nabla^2 \vec{\boldsymbol{H}} + k_c^2 \vec{\boldsymbol{H}} = 0 \end{cases}$$

$$(k_c = \omega\sqrt{\mu\varepsilon_c})$$

source region

$$\begin{cases} \nabla^2 \vec{\boldsymbol{A}} + k_c^2 \vec{\boldsymbol{A}} = -\mu \vec{\boldsymbol{J}}_{\text{int}\,roduced} \\ \nabla^2 \varphi + k_c^2 \varphi = -\dfrac{\rho_{inroduced}}{\varepsilon_c} \end{cases}$$

**Lorentz's condition**

$$\nabla \cdot \vec{\boldsymbol{A}} = -\mathrm{j}\omega\mu\varepsilon_c\varphi$$

**Field based on potential functions**

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} - \nabla \varphi$$

$$\Rightarrow$$

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -j\omega\vec{A} - j\frac{\nabla\nabla \cdot \vec{A}}{\omega\mu\varepsilon_c}$$

**Ex. Find the instantaneous value of the following field in the form of a complex vector**

$$\vec{E}(z,t)=\vec{e}_x E_{xm}\cos(\omega t-kz+\phi_x)+\vec{e}_y E_{ym}\sin(\omega t-kz+\phi_y)$$

**Solution： since**

$$\vec{E}(z,t)=\vec{e}_x E_{xm}\cos(\omega t-kz+\phi_x)+\vec{e}_y E_{ym}\cos(\omega t-kz+\phi_y-\frac{\pi}{2})$$

$$=\mathrm{Re}[\vec{e}_x E_{xm}\mathrm{e}^{\mathrm{j}(\omega t-kz+\phi_x)}+\vec{e}_y E_{ym}\mathrm{e}^{\mathrm{j}(\omega t-kz+\phi_y-\pi/2)}]$$

**So**

$$\dot{\vec{E}}_{\mathrm{m}}(z)=\vec{e}_x E_{xm}\mathrm{e}^{\mathrm{j}(-kz+\phi_x)}+\vec{e}_y E_{ym}\mathrm{e}^{\mathrm{j}(-kz+\phi_y-\pi/2)}$$

$$=(\vec{e}_x E_{xm}\mathrm{e}^{\mathrm{j}\phi_x}-\vec{e}_y \mathrm{j}E_{ym}\mathrm{e}^{\mathrm{j}\phi_y})\mathrm{e}^{-\mathrm{j}kz}$$

**Ex. The Complex vector of electric field intensity is**

$$\dot{\vec{E}}_{\mathrm{m}}(z) = \vec{e}_x \mathrm{j} E_{x\mathrm{m}} \cos(k_z z)$$

**$k_z$ and $E_{xm}$ are constant value. Find the instantaneous value of the electric field intensity**

**Solution:**

$$\vec{E}(z,t) = \mathrm{Re}[\vec{e}_x \mathrm{j} E_{x\mathrm{m}} \cos(k_z z) \mathrm{e}^{\mathrm{j}\omega t}]$$

$$= \mathrm{Re}[\vec{e}_x E_{x\mathrm{m}} \cos(k_z z) \mathrm{e}^{\mathrm{j}(\omega t + \frac{\pi}{2})}]$$

$$= \vec{e}_x E_{x\mathrm{m}} \cos(k_z z) \cos(\omega t + \frac{\pi}{2})$$

$$= -\vec{e}_x E_{x\mathrm{m}} \cos(k_z z) \sin(\omega t)$$

# Summary

- **Faraday's Law of electromagnetic induction**

$$\varepsilon_{\text{in}} = -\frac{\mathrm{d}\Psi}{\mathrm{d}t} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_S \vec{B}\cdot \mathrm{d}\vec{S} \qquad \nabla\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

- **Maxwell's Equations**

$$\begin{cases} \nabla\times\vec{H} = \sigma\vec{E} + \dfrac{\partial}{\partial t}(\varepsilon\vec{E}) \\ \nabla\times\vec{E} = -\dfrac{\partial}{\partial t}(\mu\vec{H}) \\ \nabla\cdot(\mu\vec{H}) = 0 \\ \nabla\cdot(\varepsilon\vec{E}) = \rho \end{cases}$$

- **Electromagnetic Boundary Conditions**

$$\begin{cases} \vec{e}_n\times(\vec{H}_1-\vec{H}_2) = \vec{J}_S \\ \vec{e}_n\times(\vec{E}_1-\vec{E}_2) = 0 \\ \vec{e}_n\cdot(\vec{B}_1-\vec{B}_2) = 0 \\ \vec{e}_n\cdot(\vec{D}_1-\vec{D}_2) = \rho_S \end{cases}$$

- **Wave Equations**

$$\begin{cases} \nabla^2\vec{\boldsymbol{E}} - \mu\varepsilon\dfrac{\partial^2\vec{\boldsymbol{E}}}{\partial^2 t} = 0 \\ \nabla^2\vec{\boldsymbol{H}} - \mu\varepsilon\dfrac{\partial^2\vec{\boldsymbol{H}}}{\partial^2 t} = 0 \end{cases} \qquad \begin{cases} \nabla^2\vec{A} - \varepsilon\mu\dfrac{\partial^2\vec{A}}{\partial t^2} = -\mu\vec{J} \\ \nabla^2\varphi - \varepsilon\mu\dfrac{\partial^2\varphi}{\partial t^2} = -\dfrac{\rho}{\varepsilon} \end{cases}$$

**Lorentz's gauge :** $\nabla\cdot\vec{A} + \mu\varepsilon\dfrac{\partial\varphi}{\partial t} = 0$

- **Time-Harmonic Fields**

$$\begin{cases} \nabla^2\vec{\boldsymbol{E}} + k^2\vec{\boldsymbol{E}} = 0 \\ \nabla^2\vec{\boldsymbol{H}} + k^2\vec{\boldsymbol{H}} = 0 \end{cases}$$

$$\begin{cases} \nabla^2\vec{\boldsymbol{A}} + k^2\vec{\boldsymbol{A}} = -\mu\vec{\boldsymbol{J}} \\ \nabla^2\varphi + k^2\varphi = -\dfrac{\rho}{\varepsilon} \end{cases}$$

# Homework

**7-2, 7-4, 7-7, 7-10, 7-13, 7-17, 7-18, 7-25**